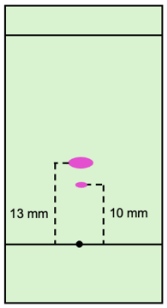
Proforma Report: Diels-Alder Reaction

Matriculation Number: A0255566U	Name: Choi Seung Eun
Reaction	
m(*N*-phenylmaleimide): 348.3 mg	V(diethyl ether): 10 mL
Crude product appearance: White crystalline particles	m(crude product): 461.8 mg
	TLC result (draw TLC plates below)
	
Column chromatography	
Eluent chosen: (ethyl acetate:hexane) = (1:2)	
V(eluent for packing & separation): 240 mL	
V(silica gel): 31 cm^3	
Height(silica gel column): 9.7 cm	
m(crude product for separation): 51.3 mg	
V(DCM for dissolving crude product): 1.0 mL	
Products	
R_f(product A): 0.371	R_f(product B): 0.286
m(product A): 22.0 mg	m(product B): 14.1 mg
Identity of product A (endo/exo): endo	Identity of product B (endo/exo): exo

NMR spectra [1. Show full spectra; 2. Insert "zoom-in" window to show the peak(s) that is/are used to confirm the endo/exo product]

Endo product (A)

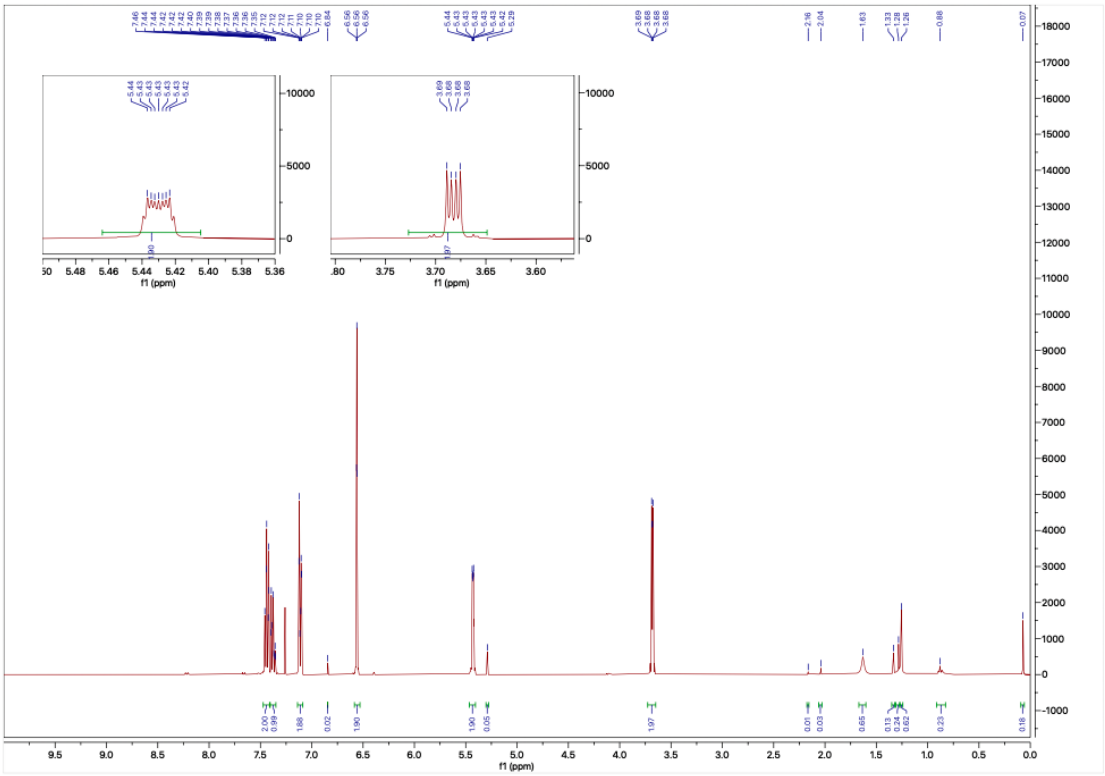


Figure 1: ^{1}H NMR spectrum of the endo product

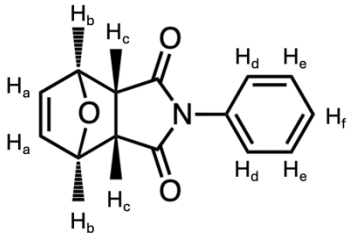


Figure 2: Structure of the endo product with protons labelled

Proton assignment	Chemical shift (ppm)	Multiplicity	Coupling constant (Hz)	Integration
H_a	6.56	t	1.0	1.90
H_b	5.44-5.41	m	-	1.90
H_c	3.68	dd	3.6 & 1.7	1.97
H_d	7.46-7.41	m	-	2.00
H_e	7.18-7.07	m	-	1.88
H_f	7.40-7.35	m	-	0.99

Table 1: Tabulation of ^{1}H NMR peaks for the endo product

Exo product (B)

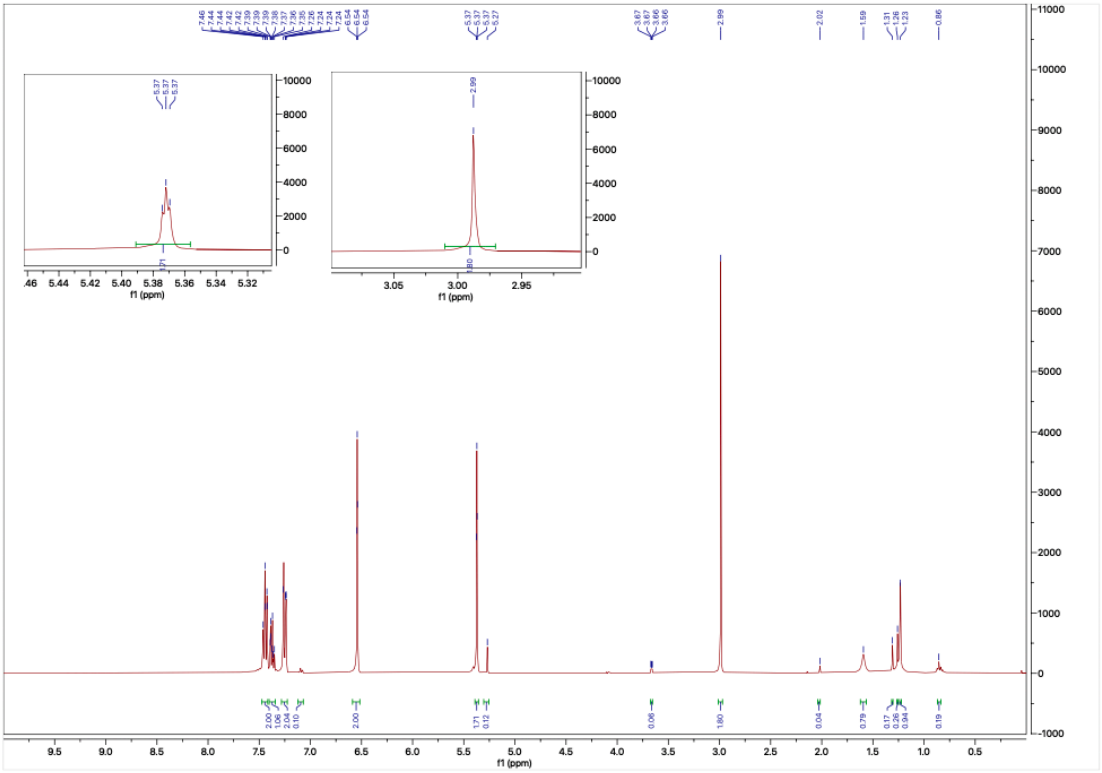


Figure 3: ^{1}H NMR spectrum of the exo product

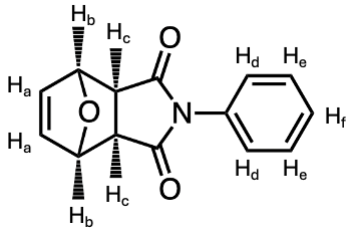


Figure 4: Structure of the exo product with protons labelled

Proton assignment	Chemical shift (ppm)	Multiplicity	Coupling constant (Hz)	Integration
H_a	6.54	t	1.0	2.00
H_b	5.37	m	1.0	1.71
H_c	2.99	s	-	1.80
H_d	7.47-7.42	m	-	2.00
H_e	7.40-7.34	m	-	2.04
H_f	7.27-7.23	m	-	1.06

Table 2: Tabulation of ^{1}H NMR peaks for the exo product